

6 (a) Define magnetic flux density.

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..... [2]

(b) Figure 6.1 shows a beam of charged particles in a device known as a mass spectrometer.

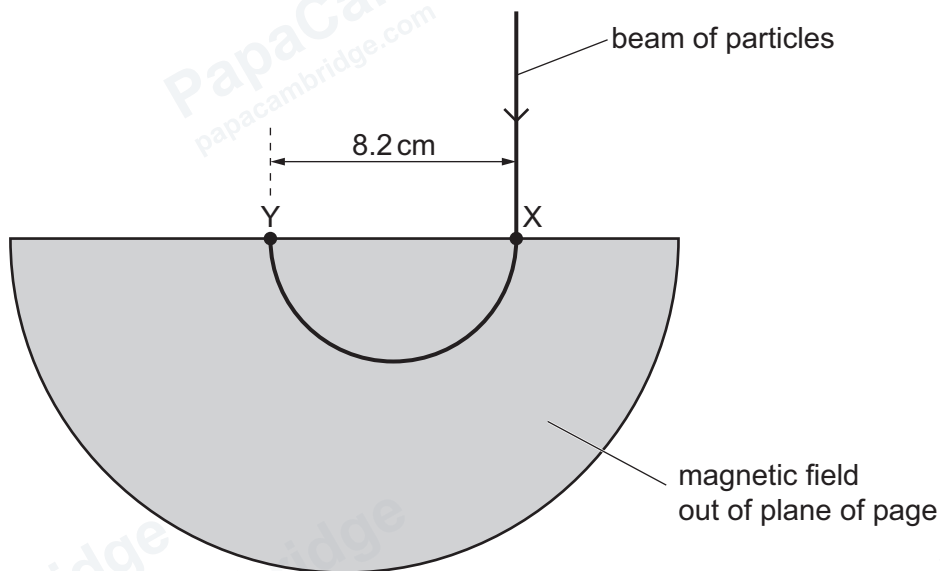


Figure 6.1

The shaded region contains a uniform magnetic field that is out of the plane of the page.

The particles enter the region of the magnetic field at point X and strike a detector at point Y.

(i) Explain whether the sign of the charge on the particles is positive or negative.

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..... [2]

- (ii) Explain why the path followed by the beam of particles inside the magnetic field is circular.

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..... [2]

- (c) Each of the particles in **6(b)** has a mass of $2.0 \times 10^{-26} \text{ kg}$ and a charge of magnitude $3.2 \times 10^{-19} \text{ C}$. All of the particles have a speed of $7.3 \times 10^4 \text{ m s}^{-1}$.

The distance XY is 8.2 cm.

- (i) Show that the centripetal acceleration of one of the particles is $1.3 \times 10^{11} \text{ m s}^{-2}$.

[2]

- (ii) Use the information in **6(c)(i)** to determine the flux density B of the uniform magnetic field. Give a unit with your answer.

$B = \dots\dots\dots$ unit $\dots\dots\dots$ [2]

- (d) The particles in **6(b)** are replaced with particles that have half of the charge and a quarter of the mass of the particles in **6(b)**. The sign of the charge and the speed of the particles are both unchanged.

On Figure 6.1, draw the path of this beam of particles. [2]

[Total: 12]